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#3,"Vikasa Bharati" , #3, Melekote, Gangasandra Road, Kalpataru Badavane, Tumakuru- 572105

Assignment Report

ON

Operating System

" Types of OS and its Features"

Submitted in partial fulfillment of 4th Semester, BCA, experimental learning

Submitted By

Student Name	Register Number
NAMRATHA L	U11SZ23S0207

Under the Guidance of

Prof. Anuradha K N
HOD,Department of. BCA
SDC, Tumakuru

2024-25

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1. ACKNOWLEDGEMENT

We take this opportunity to express our heartfelt gratitude to all those who have supported and guided us in the successful completion of this project.

Firstly, we extend we express our heartfelt gratitude to **Seshadripuram College** for its unwavering support and invaluable contributions throughout this journey.. We are especially grateful to the **Principal, Dr. Jagadeesh G.T**., for his constant encouragement and invaluable guidance, and to the **Head of the Department, Prof. Anuradha**, and also our guide, for her unwavering support and leadership, expert guidance, constant supervision, and for providing all the necessary resources and insights needed to complete this project successfully.

Additionally, we are immensely thankful to the **faculty members of the BCA Department**, whose teachings and encouragement have greatly helped us in our academic endeavors.

Our deepest thanks also go to our **parents, friends, and seniors**, whose encouragement, cooperation, and moral support have been a driving force in completing this project.

Lastly, we express our appreciation to everyone else who directly or indirectly contributed to this endeavor. Your blessings and support have been invaluable to us.

2. ABSTARCT

An operating System project summarizes the project, approach, and expected outcome. It highlights the core problem being addressed, the methods used to solve it, and the anticipated benefits or contributions of the project.

It is software that manages a computer's hardware and provide an interface for users and applications to interact with it.

The OS acts as a brige between the user and the hardware, allowing them to run programs without needing to directly control the hardware's intricacies. It also manages resources like CPU, memory, and storage to ensure efficient and fair allocation among different processes.

- ***Core Function***
  The OS is the foundation upon which all other software runs.
- ***Abstraction***
  The OS provides a high-level interface that hides the complexity of the underlying hardware.

3. INTRODUCTION

An operating system (OS) is software that acts as an intermediary between the user and the computer hardware, managing resources and providing a platform for applications to run. It's the foundation upon which all other software relies, handling tasks like managing memory, processing data, and controlling input/output devices. OSs are essential for a user-friendly and efficient computing experience.

History:

The history of operating systems (OS) can be broadly divided into eras, with each era characterized by significant advancements and shifts in how computers and software were developed and used. From the manual loading of programs in the 1940s to the development of sophisticated graphical user interfaces and mobile operating systems, OS evolution reflects the continuous push for improved efficiency, user experience, and accessibility.

- Early Beginnings (1940s-1950s)
- Rise of Unix and Personal Computers (1970s)
- Graphical User Interfaces and Networking (1980s)
- Open Source and Advanced GUIs (1990s)
- Mobile, Cloud, and AI (2000s-Present)

Significance:

operating system (OS) is crucial for any project involving computers because it acts as the foundation for all software and hardware interactions. It manages resources like CPU, memory, and storage, provides a user interface, and ensures applications can run efficiently. Without a stable and functional OS, projects relying on computer systems would be severely hampered.

- Resource Management
- User Interface
- Application Execution
- Security
- Networking
- Device Management

4.OPERATING SYSTEM

Fundamental system software that manages computer hardware and software resources, acting as an interface between the user and the computer's hardware.

How OS started:

- Operating systems (OS) begins with early batch processing systems in the 1950s, like GM-NAA I/O. These systems automated job handling for mainframe computers. Later, multitasking and time-sharing OSs emerged, like Unix, developed in 1969 at Bell Labs. The rise of personal computers saw simpler OSs like MS-DOS, and later, the introduction of graphical user interfaces (GUIs) with Windows and MacOS. Open-source OSs like Linux also gained prominence.

Features

- Process management
- Memory management
- File system management
- Device management
- User interface

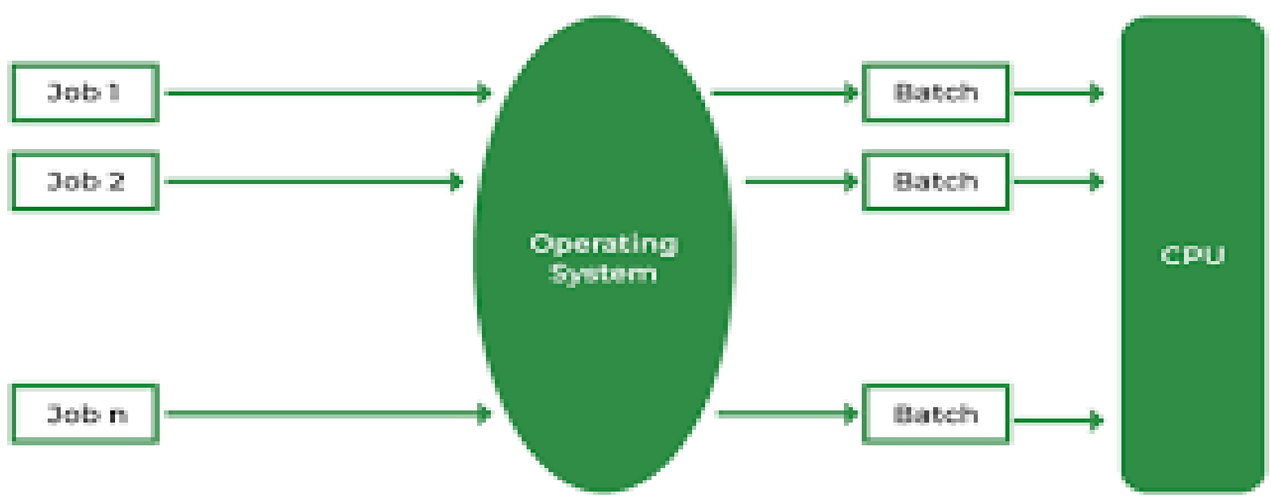
Types of OS:

1. Batch Operating System
2. Time-Sharing Operating System
3. Multiprogramming System
4. Distributed Operating System
5. Parallel System
6. Real Time Operating System

5. TYPES OF OPERATING SYSTEM

1. Batch Operating System:

A batch operating system is a system where jobs are grouped and processed together in batches, without immediate user interaction. It' s a type operating system where users submit their jobs, which are then processed one at a time, sequentially, by the operating system.



Advantages

- Scheduling is very simple
- Improved data quality
- Increased Efficiency

Disadvantages

- Higher Latency
- Manual Intervention

2.Time-sharing Operating System

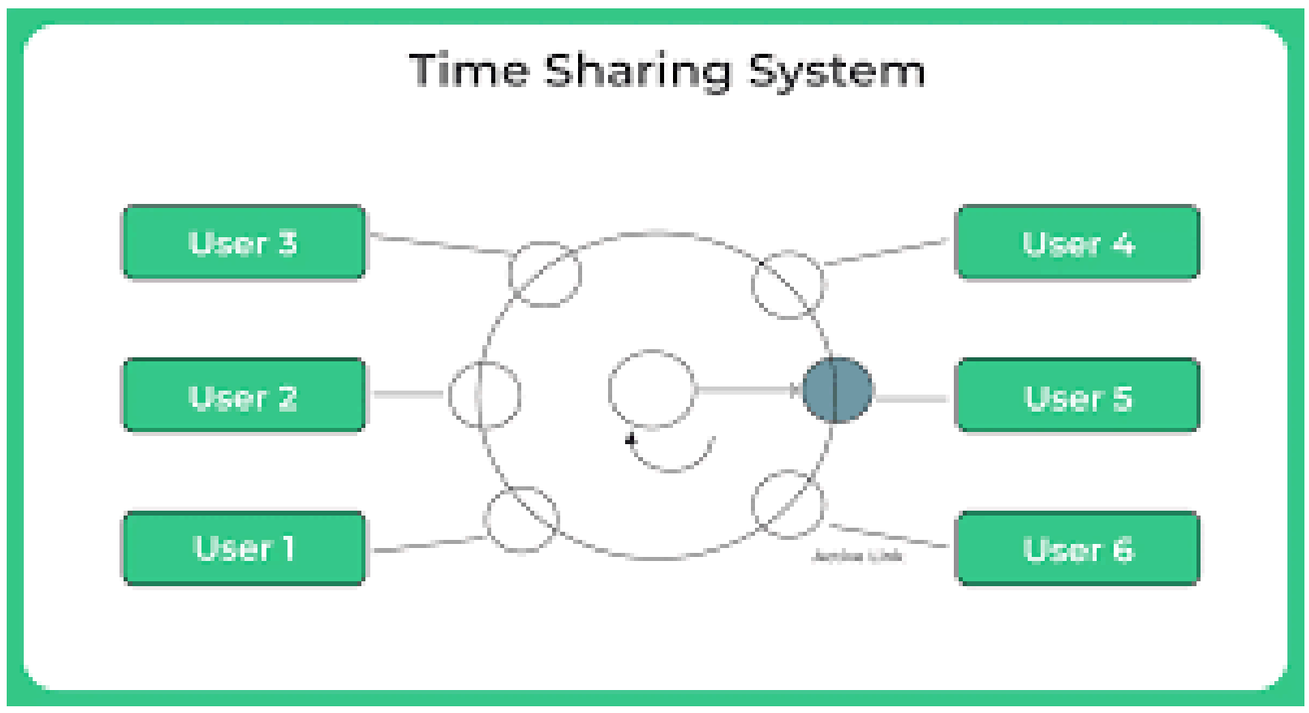
It allowes multiple users to concurrently access and use a single computer, making it appear as if each user has their own dedicated system. It achieves this by quickly switching between different users and their programs, allocating them small time slices of the CPU. This allows users to feel like they have their own individual computer, even though they are sharing the same resources.

Advantage

- Provide advantage of quick response
- Reduces CPU idle time

Disadvantages

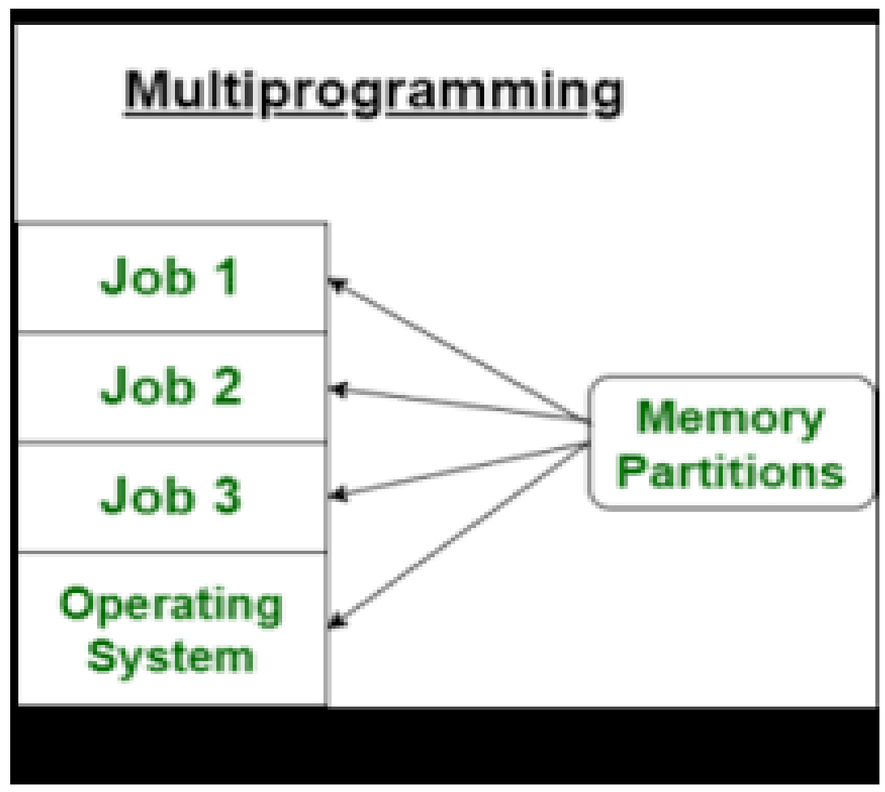
- Problem os reliability
- Data communication
- More complex



3.Multiprogramming System

It allows multiple programs to reside in memory simultaneously and share CPU time, maximizing CPU utilization.When one program waits for an I/O operation, the CPU can

switch to another ready program.This concept allows for more efficient use of computer resources compared to single-pro-gram systems.



Advantages

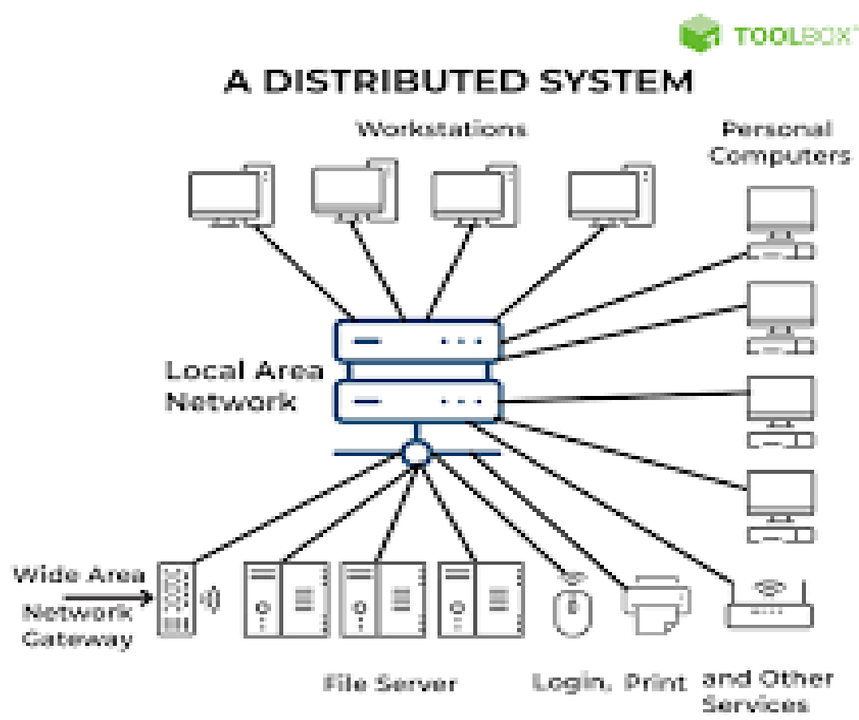
- High CPU utilization
- Shorter response time

Disadvantages

- Expensive
- Complex OS

4.Distributed Operating System

Distributed OS is a type of OS that manages and coordinates multiple computer connected via a network, making them appear as a single, unified system to users.



Advantages

- Easy data sharing

- Efficiency

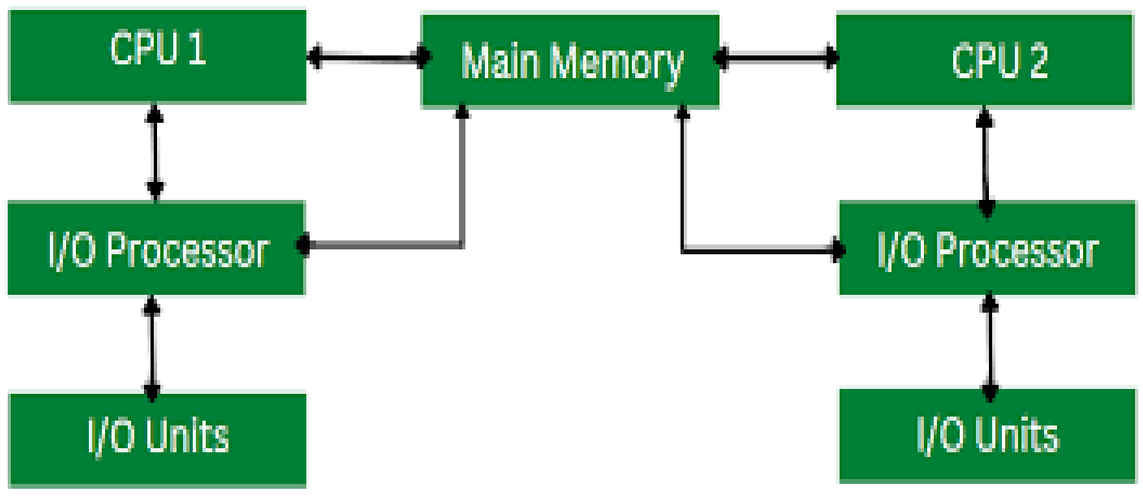
Disadvantages

- Overloading
- Data loss

5.Parallel System

A parallel System utilizers multiple processors within a single system to execute takes concurrently, improving performance and throughput, Also know as multiprocessor system.

They differ from time-sharing systems, which allow multiple users to access a single computer simultaneously, dividing CPU time for fair allocation.



Types of multiprocessor OS

- ► Asymmetric Multiprocessing
- ► Symmetric Multiprocessing

Advantages

- Increased Throughput
- Better Resources Utilization

Disadvantages

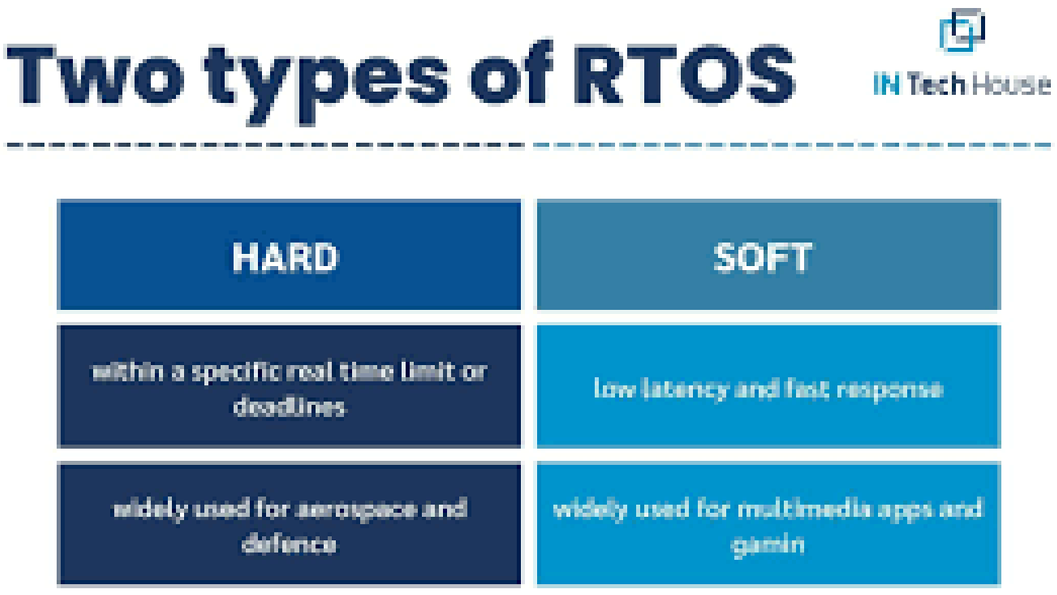
- Debugging Difficulties
- Complexity

6.Real Time Operating System

An OS that guarantees real-time applications a certain capability within a specified deadline. RTOSes are designed for critical systems and for devices like microcontrollers that are timing-specifies.The processing time requirements of an RTOS are measured in milliseconds.

Types of Real-time OS

- **Hard Real time system**
- **Soft Real time system**



Features

- ***Process management***

  The OS handles the execution of programs, allocating resources and managing their lifecycle.

- ***Memory management***

  The OS allocates and deallocates memory to different processes.
  It manages virtual memory,which allows programs to use more memory than physically available.

- ***File system management***

  The OS manages stronge devices and organizes files.
  It provides tools for accessing, manipulating, and protecting files.

- ***Device management***

  The OS manages hardware devices, providing interfaces for interacting with them.
  This includes managing I/O devices, such as keyboards, mice, and printers.

- ***User interface***

  The OS provides a graphical interface or a command-line-interface foe users to interact with the system.

6. CONCLUSION

- An Operating System is the programme that controls all other application programmes in a computer after being installed into the system first by a boot programme.

- Running system, computer operating system software that coordinate system software that coordinates data input and output, manages files, and regulates the execution of computer programmes.

- To sum up, an operating system is a piece of software that controls computer hardware and software resources and offers commom services for software applications.

- An essential component of a computer system’s system software is the operating system.

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  - **Google Chrome** – Used for browsing and researching additional details about Windows OS versions and features.
  - **Microsoft Official Website,** https://www.microsoft.com – For basic facts, version history, and Windows features.